(18)

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EE 3001: Solid-State Devices

Put down your answers in the respective places within the question paper.
All numerical answers should have proper units.

Date: 18.02.2025 Time: 50 minutes Marks: 20

1. 10^{15} phosphorus atoms/cm^3 are introduced in silicon giving rise to a donor level E_D. Assuming all phosphorus atoms are ionized and E_G = 1.12 eV, $N_C = 3.2\times10^{19}(T/300)^{3/2}$ cm^{-3}, $N_V = 1.8\times10^{19}(T/300)^{3/2}$ cm^{-3}, $n_i = 10^{10}$/cm^3 at 300K, $\tau_n = \tau_p = 1$ μsec, $\mu_n = 1350$ cm^2/V-sec, $\mu_p = 450$ cm^2/V-sec and kT = 0.026 eV at 300K, answer the following questions:

(a) The position of the Fermi level at 300 K is $E_F = E_C$ - 0.2697 eV ✓

(b) Electron concentration in the conduction band at 600 K is 2.013×10^{15}/cm^3 ✓

(c) The position of the Fermi level at 600 K is $E_F = E_C$ - 0.2785 eV ✗

(10)

If the above wafer is illuminated at 300 K so that electron-hole pairs are generated uniformly at a rate of 10^{18} cm^{-3}sec^{-1},

(d) the change in conductivity is $\Delta\sigma$ = 2.88×10^{-4} Ω^{-1}cm^{-1} ✓

e) the position of the electron Quasi Fermi Level at 300 K is $E_{Fn} = E_C$ - 0.26968 eV ✓

and the position of the hole Quasi Fermi Level at 300 K is $E_{Fp} = E_V$ + [redacted] eV

f) If the light source is suddenly switched off at $t = t_1$, then at $t = t_1 + 4$ μsec the hole concentration is 1.8315×10^{10}/cm^3 ✓

0.14365eV

[6×2 =12]

2. Consider an original silicon sample with a particular doping concentration. The sample is then doped with **additional** 5×10^{16} donor atoms/cm^3. The new sample now has a resistivity of 0.1 Ω-cm and is n-type. Assume $n_i = 10^{10}$ cm^{-3}, $\mu_n = 1350$ cm^2/V-sec and $\mu_p = 450$ cm^2/V-sec.

(4)

The type and net dopant concentration (N_D-N_A) of the original sample was P-Type / -0.3703×10^{16}/cm^3

[1+3]

3. In a uniformly doped silicon sample, the hole component of current is 10^4 times the electron component in an applied electric field. Assume $\mu_n = 1350$ cm^2/V-sec, $\mu_p = 450$ cm^2/V-sec, and $n_i = 10^{10}$/cm^3. In this sample

(a) the equilibrium electron concentration is n = 5.7735×10^7/cm^3 ✓

(b) and hole concentration is p = 1.732×10^{12}/cm^3 ✓

(4)

[2+2]

0.6846 0.26

① $N_D = 10^{15} \rightarrow$ n-type

a) $n = 10^{15} \qquad 10^{15} = 3.2\times10^{19}\, e^{-(E_c-E_F)/0.026}$

$\Rightarrow e^{(E_c-E_F)/0.026} = \ln(3.2\times10^{4}) = 10.373$

$\Rightarrow E_c - E_F = 0.2697\,eV \Rightarrow E_F = E_c - 0.2697\,eV$

b) $n_i(600) = n_i(300)\left(\frac{600}{300}\right)^{3/2} e^{-E_g/2k\,(1/600 - 1/300)} = 1.84\times10^{15}$

$n + N_A = p + N_D \Rightarrow n\cdot p = 10^{15}$

$n = 1.932\times10^{15}$

$pn = 1.8\times10^{30} \qquad ab = 1.8 \quad a-b = 1 \Rightarrow p = 0.93\times10^{15}$

c) $1.932\times10^{15} = 3.2\times10^{19}\cdot 2^{3/2}\exp\left(\frac{E_F-E_C}{0.052}\right) \Rightarrow E_C - E_F = 0.56\,eV$

d) $G_1 = 10^{18} \Rightarrow \Delta n = \Delta p = G_1 t = 10^{12}$

$\sigma_1 = n_1 q\mu_n + p_1 q\mu_p \qquad \sigma_2 = n_2 q\mu_n + p_2 q\mu_p$

$\sigma_2 - \sigma_1 = (n_2 - n_1)q\mu_n + (p_2 - p_1)q\mu_p$

$= (\Delta n q)(\mu_n + \mu_p) = 10^{12}\cdot 1.6\times10^{-19}(1800) = 2.88\times10^{-4}$

e) $n = n_i \exp\left(\frac{E_{fn} - E_i}{kT}\right)$

$10^{15} = 10^{10}\exp\left(\frac{E_{Fn} - E_i}{0.026}\right)$

$\Rightarrow E_{Fn} - E_i = 0.299 \qquad E_c - E_i = \frac{E_g}{2} = 0.56$

$\Rightarrow E_c - E_{fn} = 0.261\,eV$

$p = n_i \exp\left(\frac{E_i - E_{fp}}{kT}\right) \Rightarrow 10^{12} = 10^{10}\exp\left(\frac{E_i - E_{fp}}{0.026}\right)$

$\Rightarrow E_i - E_{fp} = 0.119 \qquad E_i = E_v + 0.56$

$\Rightarrow E_{fp} - E_v = 0.56 - 0.119 = 0.441\,eV$

f) $\Delta p(t) = \Delta p_0 e^{-t/\tau} \Rightarrow \Delta p(t) = 10^{12} e^{-4} = 1.83 \times 10^{10}$

(2) $\sigma = \frac{1}{\rho} = 10\ (\Omega cm)^{-1} = q\mu_p p + q\mu_n n$

$10 = 1.6 \times 10^{-19} \times 450\,(p + 3n)$

$\Rightarrow p + 3n = 13.9 \times 10^{16}$

$\Rightarrow n = 4.63 \times 10^{16} \qquad p = 2.15 \times 10^{3}$

$N_D + p = N_A + n \Rightarrow N_D - N_A = 4.63 \times 10^{16}$

$(N_D - N_A)_{original} = (N_D - N_A) - 5 \times 10^{16} = -0.37 \times 10^{16}$

$\rightarrow$ p-type

(3) $\frac{pq\mu_p}{nq\mu_n} = 10^4 \Rightarrow \frac{p}{n} = 3 \times 10^4$

$np = 10^{20} \Rightarrow n^2 = \frac{1}{3} \times 10^{16} = \frac{100}{3} \times 10^{14}$

$n = \frac{10}{\sqrt{3}} \times 10^7 = 5.77 \times 10^7$

$p = 10\sqrt{3} \times 10^{11} = 1.732 \times 10^{12}$